

adjustment factor. The theoretical ARMA model will cause persistence of the error term as it includes the previous day's error in the forecast, e.g., $e(t-2)$. However, our analysis has found that we could achieve more accurate estimates defining the error term only as shown above. Additionally, computation of daily volume estimates is also made easier since we do not need to maintain a series of forecast errors.

Analysis Goal

The goal of our daily volume forecasting analysis is to determine:

- Which is better ADV or MDV?
- What is the appropriate number of historical days?
- Day of week adjustment factor.
- Autoregressive volume term.

The preferred form of the ARMA model is determined via a three step process; the forecasting model should be re-examined at least on a monthly basis and recalibrated when necessary.

Step 1. Determine which is more appropriate: ADV or MDV and the historical look-back number of days.

- Compute the ADV and MDV simple forecast measure for various look-back periods, e.g., let the historical look-back period range from $t = 1$ to 30.
- Compute the percentage error between the actual volume on the day and simple forecast measure. That is:
 - $\varepsilon(t) = \ln(V(t)/\bar{V}(n))$
 - The percentage error is used to allow us to compare error terms across stocks with different liquidity.
- Calculate the standard deviation of the error term for each stock over the sample period.
- Calculate the average standard deviation across all stocks in the sample.
- Repeat the analysis for look-back periods from 1 to 30 days.
- Plot the average standard deviation across stocks for each day (from 1 to 30).

A plot of our forecast error analysis for each measure is shown in [Figure 7.3a](#) for large cap stocks and in [Figure 7.3b](#) for small cap stocks. Notice that for both large and small cap stocks the MDV measure has a lower error than the ADV. This is primarily due to the positive skew of daily volume which causes the corresponding ADV measure to be higher. Next, notice that the error term for both market cap categories follows a